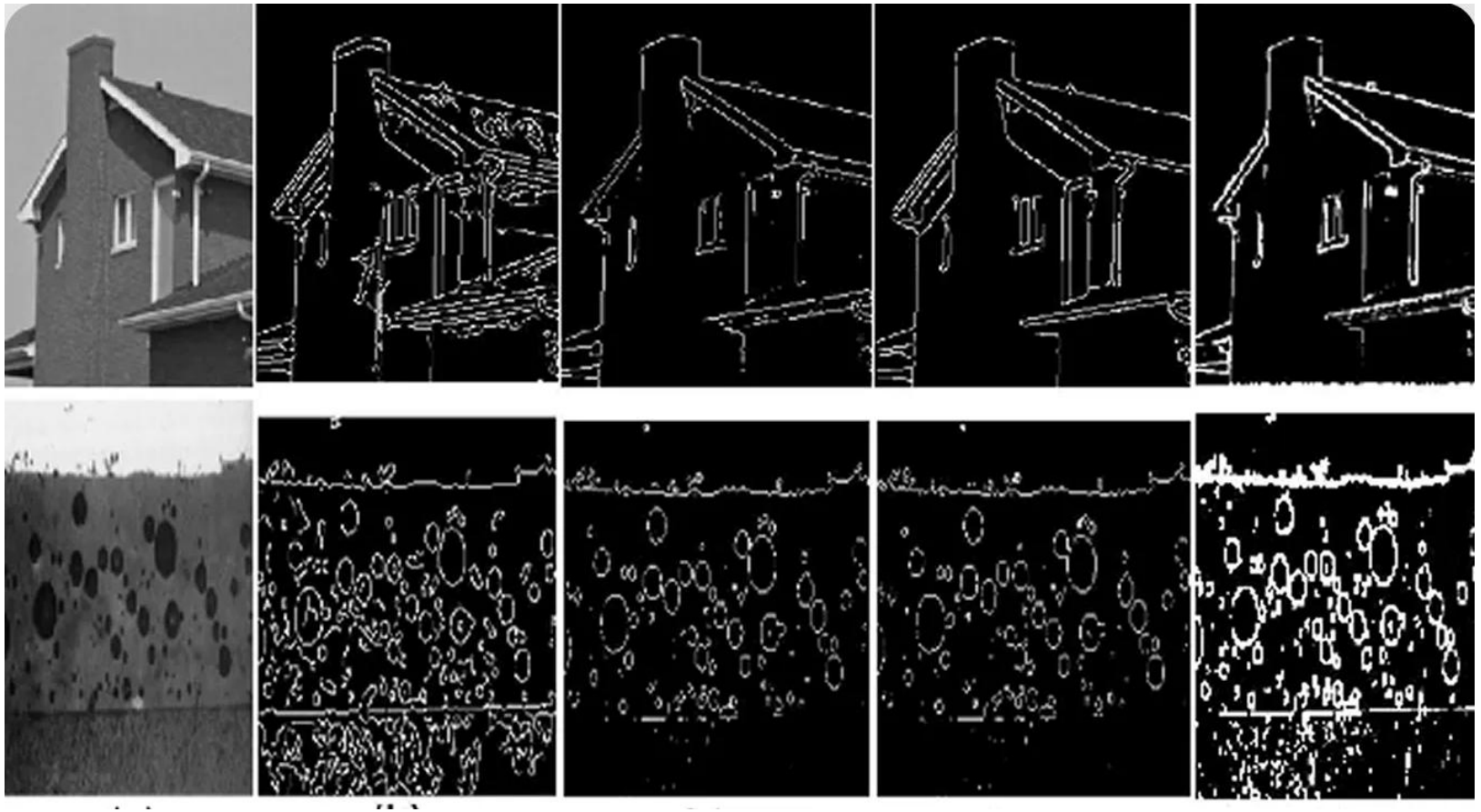
CS4186: Computer Vision

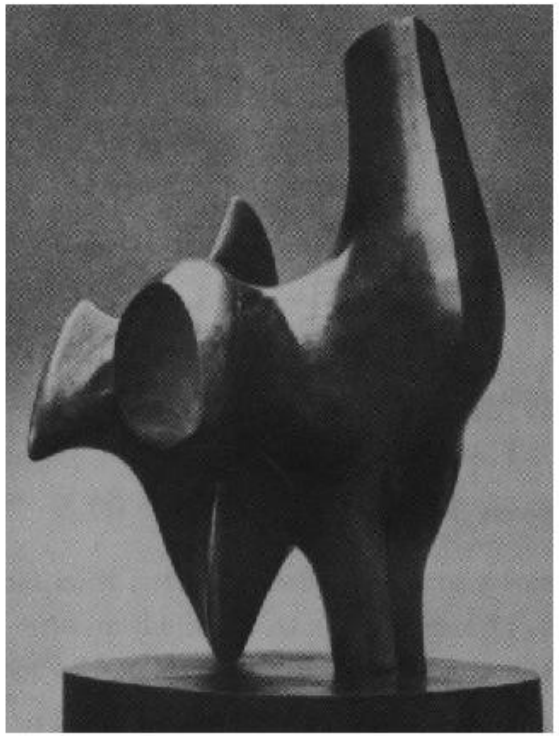
Lecture 2: Edge detection

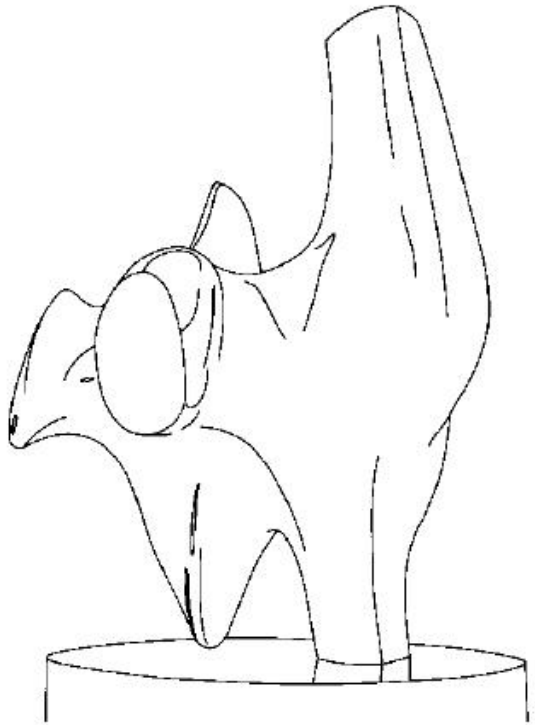
Prof. Zhu Zhiyu

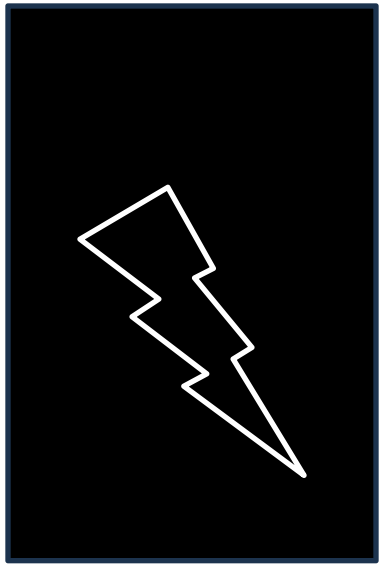
香港城市大學（東莞）
City University of Hong Kong
(Dongguan)



Edge detection

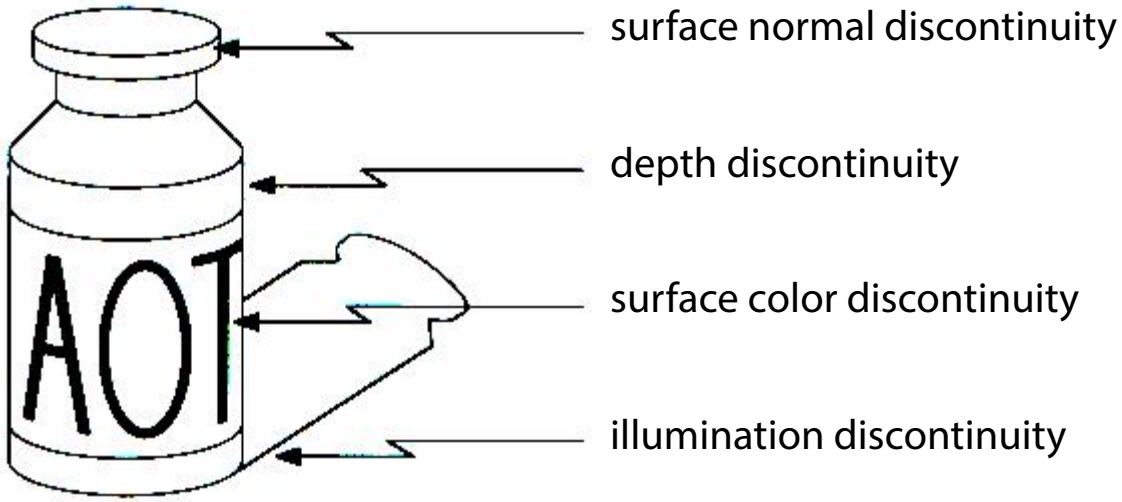






- Convert a 2D image into a set of curves
  - Extracts salient features of the scene
  - More compact than pixels

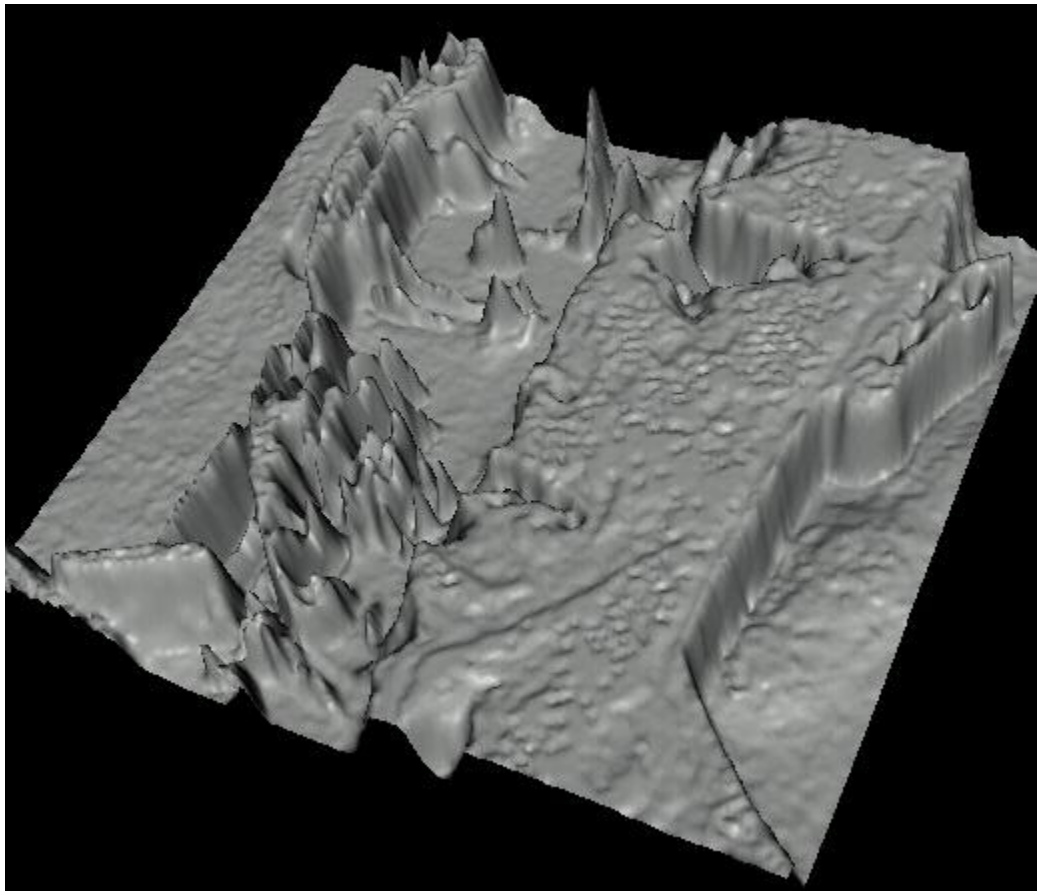
Origin of edges



- Edges are caused by a variety of factors

Images as functions…

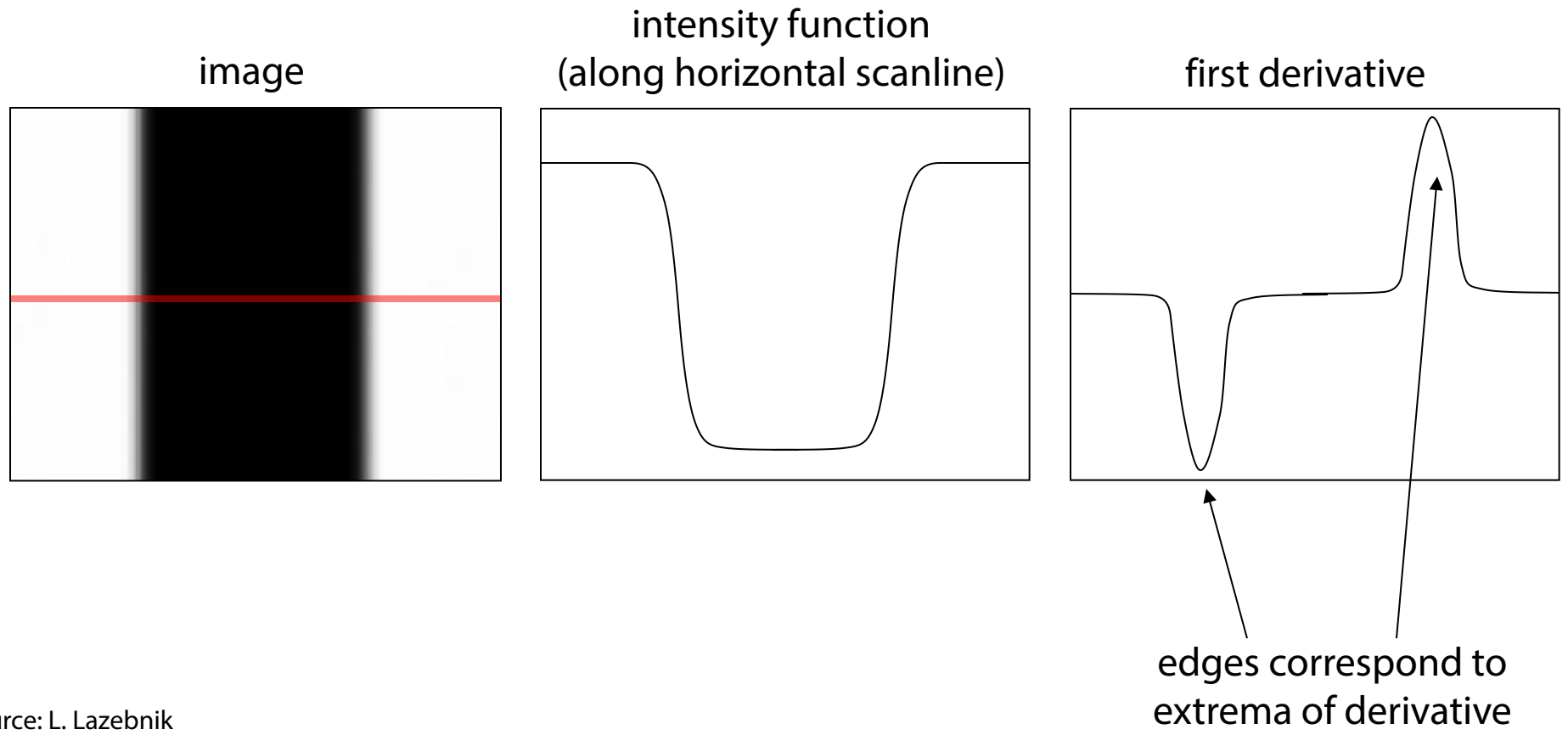




- Edges look like steep cliffs

Characterizing edges

- An edge is a place of *rapid change* in the image intensity function

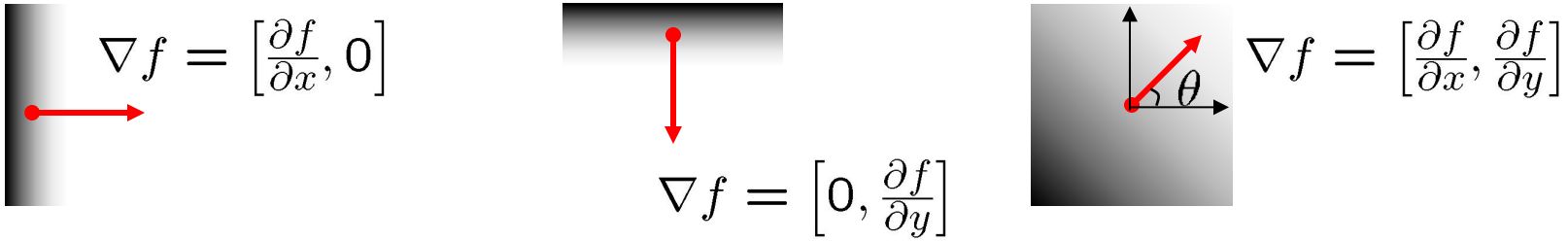


Source: L. Lazebnik

Image gradient

- The *gradient* of an image: $\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}\right]$

  The gradient points in the direction of most rapid increase in intensity



The *edge strength* is given by the gradient magnitude:

$$\|\nabla f\| = \sqrt{\left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2}$$

The gradient direction is given by:

$$\theta = \tan^{-1}\left(\frac{\partial f}{\partial y} / \frac{\partial f}{\partial x}\right)$$

- how does this relate to the direction of the edge?

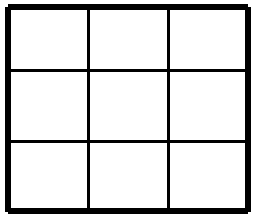
Source: Steve Seitz

Image derivatives

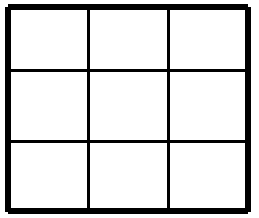
- How can we differentiate a *digital* image F[x,y]?
  - Option 1: reconstruct a continuous image, *f*, then compute the derivative
  - Option 2: take discrete derivative (finite difference)

$$\frac{\partial f}{\partial x}[x, y] \approx F[x+1, y] - F[x, y]$$

How would you implement this as a linear filter?

$\frac{\partial f}{\partial x}$: 

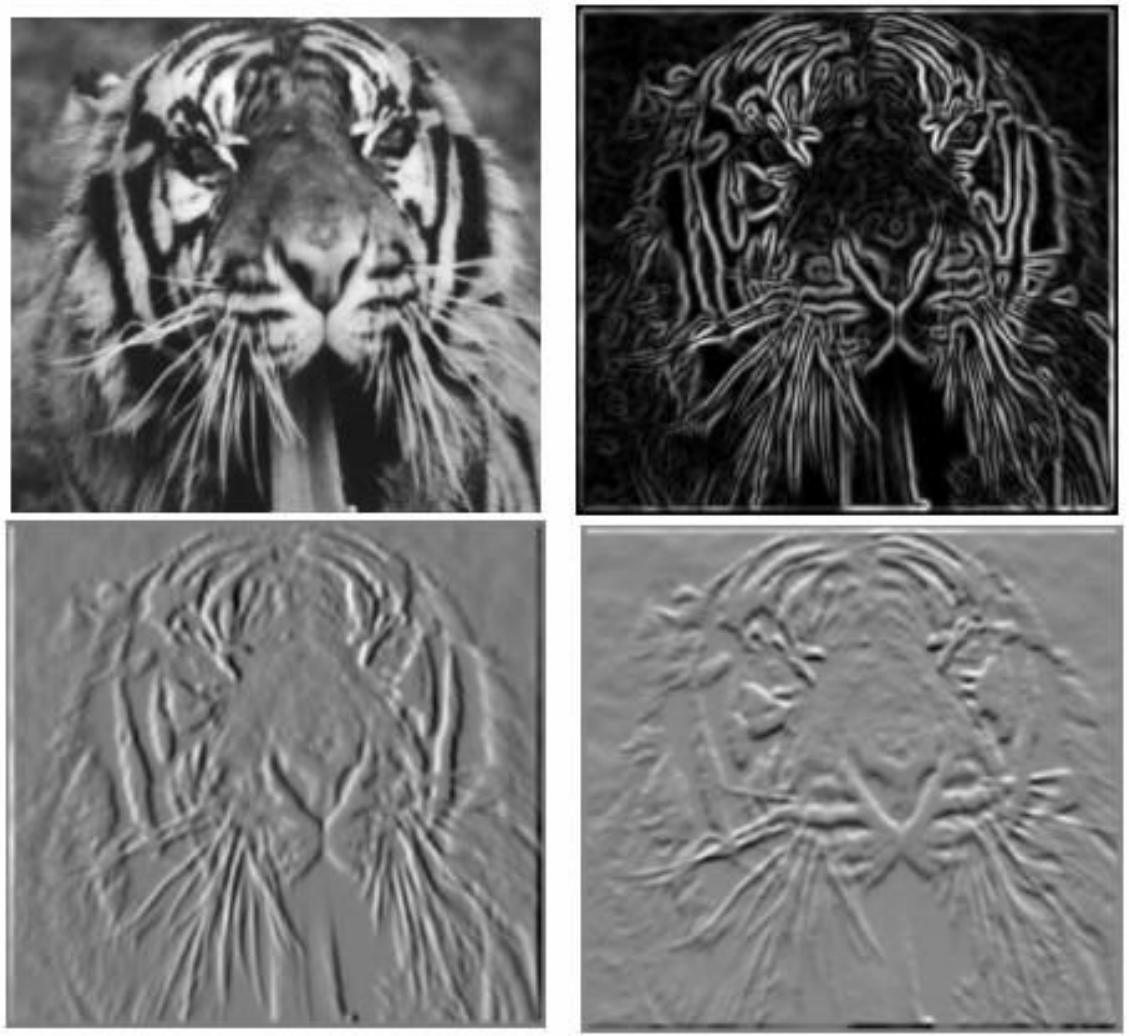
H_x

$\frac{\partial f}{\partial y}$: 

H_y

Source: S. Seitz

Image gradient



Source: L. Lazebnik